



Chapter 1

Functions and Graphs

MAE101 - CALCULUS

Ly Anh Duong

duongla3@fe.edu.vn





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- ▶ TRANSFORMATIONS: New functions from old functions
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Functions

1 Functions

A function f consists of a set of inputs, a set of outputs, and a rule for assigning **each input to exactly one output**. We denote

$$f : X \longrightarrow Y$$

$$x \longmapsto y = f(x)$$

The **set of inputs** is called the **domain** X of the function. The **set of outputs** is called the **range** Y of the function.

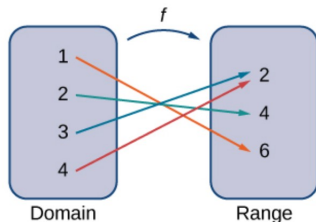
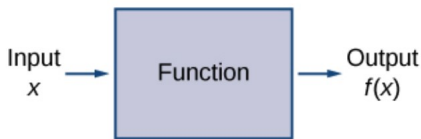


Figure: A function can be visualized as an input/output device.

Graph

1 Functions

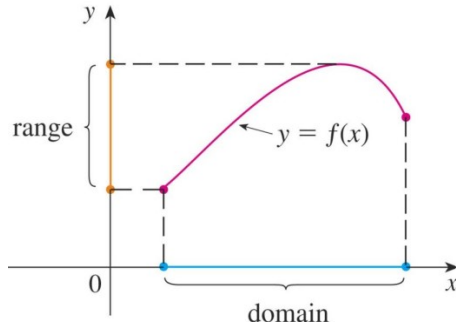
The graph of f is the **set of all points** (x, y) in the coordinate plane such that $y = f(x)$ and x is in the domain of f .

The graph of f also allows us to picture:

1. The domain of f on the x -axis.
2. Its range on the y -axis

Example. Find the the domain and the range of the functions (if it is a function).

- a. $f(n) = \sqrt{n}, \forall n \in \mathbb{N}$.
- b. $g(x)$ is any real number larger than x .



The vertical line test

1 Functions

A curve in the xy - plane is the graph of a function of x if and only if **no vertical line** intersects the curve **more than once**.

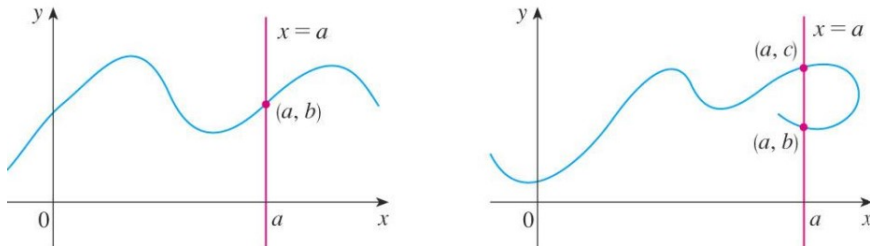


Figure: The reason for the truth of the Vertical Line Test can be seen in the figure.

SYMMETRY: Even function & Odd function

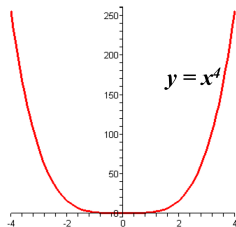
1 Functions

A function $y = f(x)$ is **even** if

$$f(x) = f(-x), \forall x \in D.$$

Geometrically, an even function is symmetrical about y-axis.

Example.

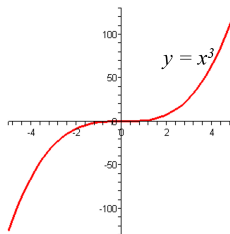


A function $y = f(x)$ is **odd** if

$$f(x) = -f(-x), \forall x \in D.$$

Geometrically, an odd function is symmetrical about the origin.

Example.



Increasing & Decreasing functions

1 Functions

A function f is called increasing on an interval I if:

$$f(x_1) < f(x_2) \text{ whenever } x_1 < x_2 \text{ in } I.$$

It is called decreasing on I if: $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ in I .

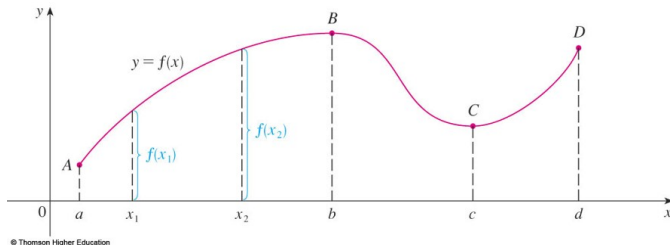


Figure: The function f is said to be increasing on the interval $[a, b]$, decreasing on $[b, c]$, and increasing again on $[c, d]$.



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2 Operations with functions

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Combinations of functions

2 Operations with functions

Two functions f and g can be combined to form new functions:

1. **Sum:** $(f + g)x = f(x) + g(x)$.
2. **Difference:** $(f - g)x = f(x) - g(x)$.
3. **Product:** $(fg)(x) = f(x)g(x)$.
4. **Quotient:** $\frac{f}{g}(x) = \frac{f(x)}{g(x)}, g(x) \neq 0$.

Example. Given the functions $f(x) = 2x - 3$, $g(x) = x^2 - 1$, find each of the following functions and state its domain.

a. $(f + g)x$

b. $(f - g)x$

c. $(fg)(x)$

d. $\frac{f}{g}(x)$

Combinations of functions

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- | | |
|---------------|---------------------|
| a. $(f + g)x$ | b. $(f - g)x$ |
| c. $(fg)(x)$ | d. $\frac{f}{g}(x)$ |

Solution.

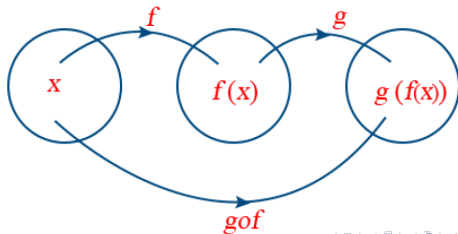
- a. $(f + g)x = f(x) + g(x) = x^2 + 2x - 4$. The domain $D = \mathbb{R}$.
- b. $(f - g)x = f(x) - g(x) = -x^2 + 2x - 2$. The domain $D = \mathbb{R}$.
- c. $(fg)(x) = f(x)g(x) = 2x^3 - 3x^2 - 2x + 3$. The domain $D = \mathbb{R}$.
- d. $\frac{f}{g}(x) = \frac{f(x)}{g(x)} = \frac{2x - 3}{x^2 - 1}$. The domain $D = \mathbb{R} \setminus \{-1, 1\}$.

Function composition

2 Operations with functions

Consider the function $f : A \longrightarrow B$, and the function $g : D \longrightarrow E$ with domain D and range E . If $B \subset D$, then the composite function $(g \circ f)(x)$ is the function with domain A such that

$$(g \circ f)(x) = g(f(x))$$



Example. Given

$$f(x) = x^2 + 1, g(x) = 1/x.$$

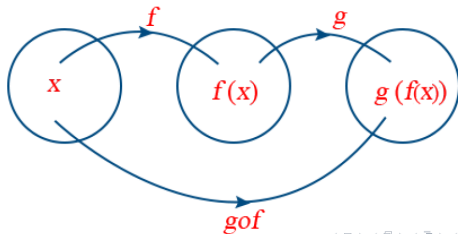
1. Find $(g \circ f)(x)$, $(f \circ g)(x)$ and state its domain and range.
2. Evaluate $(g \circ f)(4)$, $(g \circ f)(-1/2)$.
3. Evaluate $(f \circ g)(4)$, $(f \circ g)(-1/2)$.

Function composition

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Solution.

$(g \circ f)(x)$: $D = \mathbb{R}$, the range is $(0, 1]$; $(f \circ g)(x)$: $D = \mathbb{R} \setminus \{0\}$, the range is $(1, +\infty)$



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3 TRANSFORMATIONS: New functions from old functions

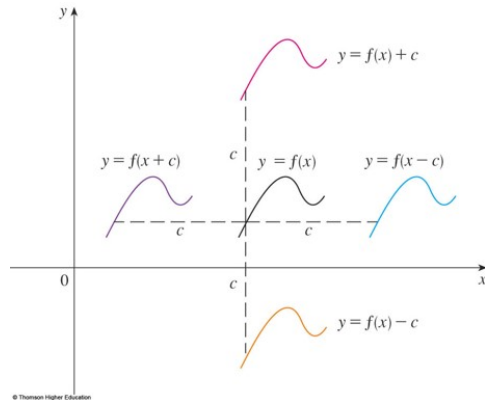
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Vertical shift & Horizontal shift

3 TRANSFORMATIONS: New functions from old functions

Suppose $c > 0$

1. To obtain the graph of $y = f(x) + c$, shift the graph of $y = f(x)$ a distance c units **upward**.
2. To obtain the graph of $y = f(x) - c$, shift the graph of $y = f(x)$ a distance c units **downward**.
3. To obtain the graph of $y = f(x - c)$, shift the graph of $y = f(x)$ a distance c units to the **right**.
4. To obtain the graph of $y = f(x + c)$, shift the graph of $y = f(x)$ a distance c units to the **left**.



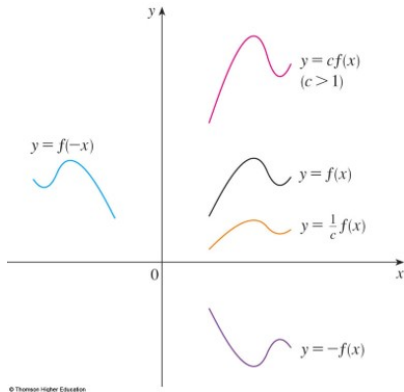
Note: How about the case $c < 0$?

Vertical & Horizontal stretch and compression

3 TRANSFORMATIONS: New functions from old functions

Suppose $c > 1$

1. To obtain the graph of $y = cf(x)$,
stretch the graph of $y = f(x)$ vertically
by a factor of c .
2. To obtain the graph of $y = \frac{1}{c}f(x)$,
compress the graph of $y = f(x)$
vertically by a factor of c .
3. To obtain the graph of $y = f(cx)$,
compress the graph of $y = f(x)$
horizontally by a factor of c .
4. To obtain the graph of $y = f(\frac{1}{c}x)$,
stretch the graph of $y = f(x)$
horizontally by a factor of c .



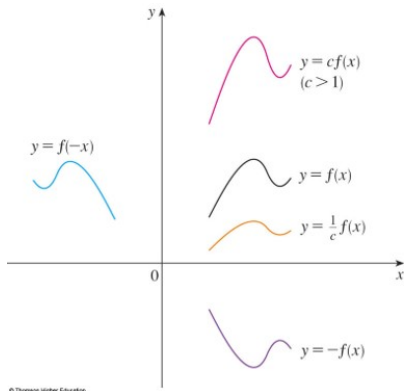
Note: How about the case $0 < c < 1$?

Reflections

3 TRANSFORMATIONS: New functions from old functions

1. To obtain the graph of $y = -f(x)$,
reflect the graph of $y = f(x)$ about
the x -axis.

2. To obtain the graph of $y = f(-x)$,
reflect the graph of $y = f(x)$ about
the y -axis

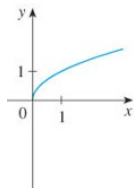


Example 1

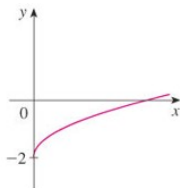
3 TRANSFORMATIONS: New functions from old functions

Label the following graph from the graph of the function $y = \sqrt{x}$ shown in the part

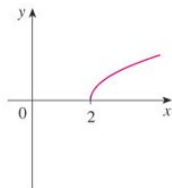
(a): $y = f(x) - 2$, $y = f(x - 2)$, $y = -f(x)$, $y = 2f(x)$, $y = f(-x)$



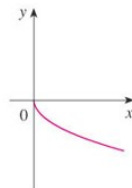
(a) $y = \sqrt{x}$



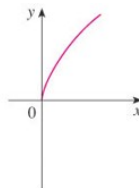
(b) $y = \sqrt{x} - 2$



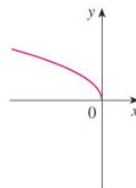
(c) $y = \sqrt{x - 2}$



(d) $y = -\sqrt{x}$



(e) $y = 2\sqrt{x}$



(f) $y = \sqrt{-x}$

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Example 2

3 TRANSFORMATIONS: New functions from old functions

The figure illustrates these stretching transformations when applied to the **cosine function** with $c = 2$.

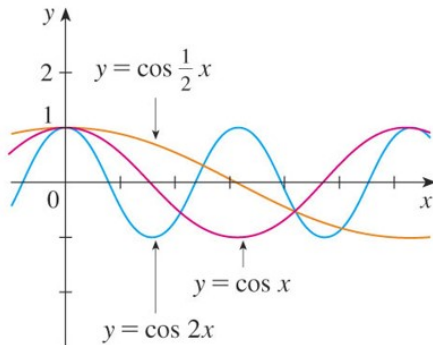
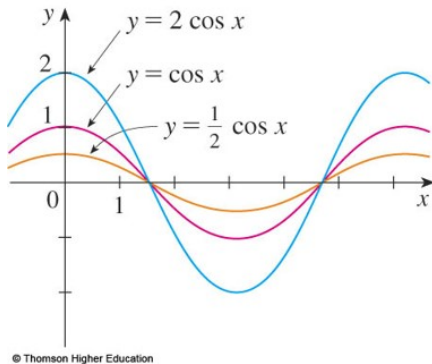




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Problems

4 Problems

1. Find the domain of each function:

a) $f(x) = \sqrt{x+2}$ b) $f(x) = \frac{1}{x^2 - x}$ c) $f(x) = \ln(x+1) - \frac{x}{\sqrt{x-1}}$

2. Find the range of each function:

a) $f(x) = \sqrt{x-1}$ b) $f(x) = x^2 - 2x$ c) $f(x) = \sin x$

3. Determine whether is even, odd, or neither

a) $f(x) = x^3 - 2015$ b) $f(x) = \sin(2x^4)$ c) $f(x) = \frac{x}{x^2 + 1}$.

4. Let $f(x) = \sqrt{x+1}$ and $g(x) = \sqrt{5+x}$. Find $g \circ f$?

Problems

4 Problems

5. Use the table to evaluate the expression $(f \circ g)(2)$:

x	1	2	3	4	5	6
$f(x)$	3	4	2	0	1	2
$g(x)$	3	6	1	3	4	0

6. Let $f(x) = \frac{x^2 + x + 1}{x}$. Find

a) $f\left(x + \frac{1}{x}\right)$ b) $f(2x - 1)$

7. Explain how the following graphs are obtained from the graph of $f(x)$

a) $f(x - 4)$ b) $f(x) + 3$ c) $f(x - 2) - 3$



Q&A

Thank you for listening!